

MQA (with KV-caching)

① Dim

$$\bullet \boxed{X_{\text{new}} \in \mathbb{R}^{(S_{\text{new}}, d_{\text{model}})}}$$

• S_{context} = length of past tokens in the KV-cache.

• S_{new} = number of new tokens being generated.

• d_{model} = embedding dimension.

• h = number of query heads.

• $d_k = d_m / h$ = dim per head.

→ we will compute Q and use cached K, V and compute attention.

② Compute Q, K, V (but cache K, V)

1) Compute Q for new tokens

Since K, V are cached, we only compute Q for new tokens:

$$\boxed{Q = X_{\text{new}} \cdot W_Q}, \quad W_Q \in \mathbb{R}^{(d_m, d_m)}$$

• ~~Q shape (d_m, d_m)~~

$$(S_{\text{new}}, d_m) \times (d_m, d_m) = (S_{\text{new}}, d_m)$$

$$\boxed{Q \in \mathbb{R}^{(S_{\text{new}}, d_m)}}$$

$$\text{Comput. cost} = O(S_{\text{new}}, d_m) \cdot O(d_m, d_m) = \boxed{O(S_{\text{new}} \cdot d_m^2)}$$

$$\text{Memory cost} = \boxed{O(S_{\text{new}}, d_m)}$$

2) Compute K, V for new tokens and appends to Cache

→ we still compute K, V once per token and store them in cache

$$K_{\text{new}} = X_{\text{new}} \cdot W_K, \quad V_{\text{new}} = X_{\text{new}} \cdot W_V$$

$$W_K \in \mathbb{R}^{(d_m, d_k)}$$

$$W_V \in \mathbb{R}^{(d_m, d_v)}$$

$$(S_{\text{new}}, d_m) \times (d_m, d_k) = (S_{\text{new}}, d_k)$$

Final shape:

$$K_{\text{new}}, V_{\text{new}} \in \mathbb{R}^{(S_{\text{new}}, d_k)}$$

→ we append them to KV cache:

$$K_{\text{cache}} = [K_{\text{context}}, K_{\text{new}}] \in \mathbb{R}^{(S_{\text{total}}, d_k)}$$

$$V_{\text{cache}} = [V_{\text{context}}, V_{\text{new}}] \in \mathbb{R}^{(S_{\text{total}}, d_v)}$$

$$\text{where } S_{\text{total}} = S_{\text{context}} + S_{\text{new}}$$

$$\text{Comp. Cost: } O(S_{\text{new}} \cdot d_m \cdot d_k)$$

$$\text{Memory: } K_{\text{cache}}, V_{\text{cache}} \rightarrow O(S_{\text{total}}, d_k)$$

③ Reshape Q into multiple Heads

→ we reshape Q into multiple query heads

$$Q \rightarrow (S_{\text{new}}, h, d_k) \xrightarrow{T} (h, S_{\text{new}}, d_k)$$

$$Q \in \mathbb{R}^{(h, S_{\text{new}}, d_k)}$$

④ Compute Attention Scores A

$$A = \frac{Q \cdot K_{\text{cache}}^T}{\sqrt{d_k}}$$

$$(h, S_{\text{new}}, d_k) \times (h, d_k, S_{\text{total}}) = (h, S_{\text{new}}, S_{\text{total}})$$

$$\text{Comp-Cost} = O(S_{\text{new}} \cdot S_{\text{total}} \cdot d_m)$$

$$\text{Memory-Cost} = \text{for } A \rightarrow O(h \cdot S_{\text{new}} \cdot S_{\text{total}})$$

$$\text{⑤ apply softmax, } A \in \mathbb{R}^{(h, S_{\text{new}}, S_{\text{total}})}$$

⑥ Compute output O

$$O = A V_{\text{cache}}$$

$$(h, S_{\text{new}}, S_{\text{total}}) \times (h, S_{\text{total}}, d_v) = (h, S_{\text{new}}, d_v)$$

$$O \in \mathbb{R}^{(h, S_{\text{new}}, d_v)}$$

Comput-cost:

$$O(S_{\text{new}} \cdot S_{\text{total}} \cdot d_m)$$

Memory:

$$O(S_{\text{new}} \cdot d_m) \quad \text{for } O$$

D Concatenated Heads of Project W/O

$$O \in \mathbb{R}^{(S_{nw}, d_m)}$$

$$\text{final proj} : (S_{nw}, d_m) \times (d_m, d_m) \rightarrow (S_{nw}, d_m)$$

$$\text{Computational Cost} = O(S_{nw} \cdot d_m^2)$$